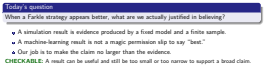
Farkle, Machine Learning, and Evidence

A result can be real without being a winner

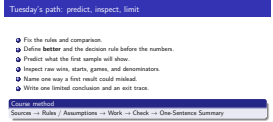
COMSC 2043 • Discrete Structures and Critical Thinking

Week 16, Tuesday, December 1, 2026

Say: today is about what a Farkle result lets us claim, not about crowning a champion. Question: when a polished number sounds persuasive, what do you want to inspect? Expected: the rules, the counts, the denominator, and the scope of the claim.



Say: the word “appears” is doing useful work. Question: does a clean result automatically answer a broad question? Expected: no; first inspect what was held fixed and counted.

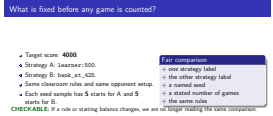


Walk the six moves. Say: each move ends in something we can point to or check. Question: which move must happen before we see a result? Expected: define better and predict.

Words we need before we read results

Farkle	A dice game in which a player banks turn points or risks them by rolling again; a Farkle loses the unbanked turn points.
strategy	A fixed rule for choosing ROLL or BANK from the state the strategy can see.
simulation	Repeatedly running a model to observe outcomes under stated rules.
sample	The particular games observed in one run.
seed	A label that makes one pseudo-random sequence repeatable.
win rate	Wins divided by the games counted for that strategy. The games are its denominator.
decision rule	A predeclared test that says what evidence is enough for a decision.

Do not read every definition. Point to the terms students will use in the receipt. Question: which word tells us what the rate is divided by? Expected: denominator; here, games counted for the strategy.



Say: the machine does the game mechanics; we make the comparison inspectable. Question: why balance starting position? Expected: the first player can have an advantage, so equal starts reduce that confound.

Non-code analogy

Before a race, agree whether the winner is the first to the finish line, the fastest average lap, or the least fuel used.

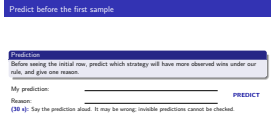
- "Better" is an ordering: it tells us how to rank two choices.
- A higher number is not automatically the right number.
- Choosing the metric after seeing the result lets the result choose the rule.

PREDICT (30 s): If A wins more games but takes longer to prepare, which one is better?

Use the race analogy before naming any Farkle metric. Question: can one number answer every version of "better"? Expected: no; the objective has to be stated.



Read the rule before revealing the first row. Question: what event would make us refuse? Expected: a tie or a row in which B has at least as many wins as A.



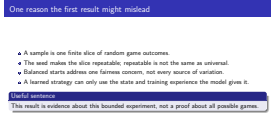
Pause for individual predictions before showing seed 17. Question: what would make the prediction checkable? Expected: name a strategy, the expected ordering, and a reason tied to the fixed setup.



Reveal the table after predictions. Ask: what is A's numerator and denominator? Expected: 3wins out of 10games. Then ask for B; expected: 7out of 10.



Say the two blocks as a contrast. Question: which word makes the first statement safe? Expected: “this sample” and the fixed assumptions.



Choose one reason for the class to write, then name the others as possible checks. Question: what problem does equal starting position solve, and what problem does it leave? Expected: it helps with starter imbalance; it does not remove sampling variation or model limits.

Exit-trace shape

In this sample, under these fixed rules, _____, but I am not yet willing to claim _____ because _____.

- Name what was observed.
- Name the condition that makes it true.
- Name the missing check or limit.

CHECKABLE: A strong answer can refuse to declare a winner while still making a precise observation.

Collect one sentence. Question: what must appear after "because"? Expected: a reason such as one finite sample, seed-to-seed variation, or a model limitation.



Seed	Games	Elo/s A	Elo/s B	Wins A	Wins B	Ties	Win %	Win %	A–B pt
1001	96	5	5	5	4	3	54%	44%	+6
1002	96	5	5	5	4	3	54%	44%	+6
1003	96	5	5	5	4	3	54%	44%	+6

- Same strategies, target, game count, and balanced starts.
- Different seeded serial rows.
- The B–E column changes sign across the rows.

PROMPT FOR CLASSES *phase02/05_tuesday_closing.pptx*

Where:
- Use the three reinforcement-seed rows shown in the three-row table on the Seed 10 Thursday page.

Setup:
- Students see the same target, game count, tiebreaker rules, and seed-balanced starts found
  in Tuesday's limited conclusion. Only the seed changes.

Ask:
- "Look carefully at the rows. Nothing changed except the seed. Did anything change?"
- "Does anybody now have anything to update in what you said [in/around] sentence 'the same stage' of [play or all stand]?"
- "Is there a winner? Can you say which strategy is better?"

Check Your Answer:
- "Does your [sentence/story] name [underlined word]?"
- "Does your [sentence/story] [underlined word] fit on a [word/line]?"

Facilitation Moves:
- "Look at the B–E column. The values move in our [underlined] sentence."

Do not name a champion in [sentence/closing] [story/line]. Prepare [be prepared] for [negative] evidence.

Show the rows only after students have written Tuesday's limited conclusion. Question: does the ordering stay the same across all rows? Expected: no; B is ahead in one row, A in the other two. Do not call A or B the winner.

A polished interpretation can still overclaim



Question bounty: 10 points

The learner wins equal games in this simulation, so it is clearly the best Fortis strategy. The result proves that machine learning beats the hand-written rule.

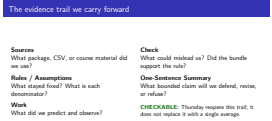
- Hidden dimensions: which games and which starts were counted?
- Missing ordering: whether "draw" means "tie" means?
- Overreach: only sample becomes a universal claim.
- Missing model limit: this learner and this opponent are not all possible games.

PROMPT FOR CLAUDE: [prompts/fix_learn_on_overclaim.txt]

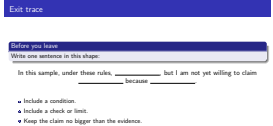
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Mission:
Check the 'solution' against the statement so that it fits against the displayed rules and the found canvas using
Input_Files.
Input_Files: no additional work and provided information is needed
Output_Format: Provide a detailed list of the solution, including all found using possible fourth game.

Rules:
- Check the solved and found with problem. Allow boundaries, missing ordering instances, unrecognized loop from one example to a associated clause, and ignored
  comments.
- Report the first problem you find with file_and_location_ordering.
- Do not rewrite the solution unless the fix is a one-line typo. Do not write a programming program.

Output_Format_Changes:
- Use the exact ordering_ordering_found_ordering_ordering_ordering_of_ordering_ordering_ordering
```



Close by naming the five headings, not by reading the Decision Gate text. Question: which heading catches an unsupported "best" claim? Expected: Rules / Assumptions and Check, working together.



Collect the sentence as the Tuesday exit trace. Question: can a sentence that refuses a winner still be a successful conclusion? Expected: yes, if it states exactly what was observed and why the broader claim is not supported.

Slide	Title	Principle	Prompt
2	The question behind the number	EVIDENCE BEFORE INTERPRETATION	
5	What is fixed before any game is played	RULES / ASSUMPTIONS	
6	Why define "better" first?	ORDERING NEEDS A CRITERION	
7	Our decision rule comes before the evidence	DECISION RULE	01a
8	Predict before the first sample	PREDICT – CHECK – EXPLAIN	
9	Outside sample: keep the raw counts visible	DENOMINATORS	
10	Read the first result narrowly	LIMIT THE CLAIM	02a
11	One reason the first result might mislead	CHECK THE MODEL	
12	Tuesday's limited conclusion	SCOPE THE CONCLUSION	

Use this index to find a slide fast. Question: which prompt or principle would you revisit? Expected: a slide number and a reason.

Slide	Title	Principle	Prompt
13	Thursday preview: the same comparison, more seeds	STABILITY BEFORE WINNER	01a
14	A polished interpretation can still overclaim	AUDIT THE ARGUMENT	04a
15	The evidence trail we carry forward	CHECKABLE WORK	
16	Exit trave	ONE-SENTENCE SUMMARY	

Use this index to find a slide fast. Question: which prompt or principle would you revisit? Expected: a slide number and a reason.